

Hybrid Cloud Infrastructure Runbook

Azure Virtual Network Gateway (VNG) to On-Premises Linux strongSwan VPN (Fully Sanitized)

SYSTEM VERIFIED & OPERATIONAL

1. Network Topology & IP Allocation

Azure Environment (Hub)	On-Premises Gateway (Branch)
• Region: Sweden Central • VNet CIDR: 10.0.0.0/16 • Gateway Subnet: 10.0.255.0/27 • Target VPN IP: <AZURE_VNG_PIP> • BGP ASN: 65515 • BGP Peer IP: 10.0.255.30	• OS: Ubuntu 24.04 LTS (strongSwan + FRR) • Local Subnet: 192.168.0.0/16 • Router Private IP: 192.168.254.4 • Router Public IP: <SWAN_PIP> • BGP ASN: 65001 • BGP Peer IP: 192.168.254.4

2. Azure Infrastructure Configuration (IaC)

The following resources were successfully provisioned and configured in the Azure portal via ARM templates to establish the cloud-side of the tunnel.

Virtual Network Gateway (VNG)	
Gateway Type	Vpn
VPN Type	RouteBased
SKU	VpnGw1AZ (Generation 2)
BGP Enabled	True (ASN: 65515)

Local Network Gateway (LNG)

Gateway IP Address	<SWAN_PIP> (On-Premises Public IP)
Address Space	192.168.0.0/16 (Fallback for BGP propagation)
BGP Enabled	True (ASN: 65001, Peer: 192.168.254.4)

Connection Object

Connection Type	IPsec / IKEv2
Authentication	Pre-Shared Key (PSK)
BGP Enabled	True
DPD Timeout	45 Seconds

Network Security Group (NSG) Rules

Allow-IPSec-IKE	Inbound / UDP 500 / Source: <AZURE_VNG_PIP>
Allow-IPSec-NATT	Inbound / UDP 4500 / Source: <AZURE_VNG_PIP>
Allow-BGP	Inbound / TCP 179 / Source: 10.0.255.30
Allow-SSH	Inbound / TCP 22 / Source: 192.168.0.0/16

3. On-Premises Router Configuration (strongSwan + FRR)

The Linux gateway required precise cryptographic matching and routing protection to establish the connection with the Azure VNG.

```
/etc/ipsec.conf
```

```
config setup
    charondebug="ike 2, knl 2, cfg 2"
    uniqueids=no

conn %default
    ikelifetime=1440m
    keylife=60m
    rekeymargin=3m
    keyingtries=%forever
    keyexchange=ikev2
    authby=secret
    dpdaction=restart
    dpddelay=30s
    dpdtimeout=120s

conn azure-s2s
    left=192.168.254.4
    leftid=<PIP_STRONGSWAN_ROUTER>
    leftsubnet=192.168.0.0/16
    right=<AZURE_VNG_PIP>
    rightid=<AZURE_VNG_PIP>
    rightsubnet=10.0.0.0/16
    ike=aes256-sha256-modp2048,aes256-sha1-modp1024!
    esp=aes256-sha256,aes256-sha1!
    auto=add
```

```
/etc/strongswan.conf
```

```
charon {
    load_modular = yes
    # CRITICAL: Prevents hijacking of management routes
    install_routes = no
    plugins {
        include strongswan.d/charon/*.conf
    }
}
include strongswan.d/*.conf
```

```
/etc/frr/frr.conf
```

```
frr version 8.1
frr defaults traditional
hostname strongswan-router
log syslog informational
no ipv6 forwarding
service integrated-vtysh-config
!
router bgp 65001
 neighbor 10.0.255.30 remote-as 65515
 neighbor 10.0.255.30 ebgp-multihop 255
 neighbor 10.0.255.30 update-source 192.168.254.4
!
address-family ipv4 unicast
 network 192.168.0.0/16
 neighbor 10.0.255.30 activate
exit-address-family
!
line vty
!
```

4. Protocol Debugging & Troubleshooting History

The following edge-cases were encountered and successfully resolved during deployment:

1. Linux Kernel Route Hijacking (SSH Lockout)

Diagnosis: Upon daemon startup, strongSwan auto-injected a default policy route (`0.0.0.0/0`) into XFRM table 220, intercepting management TCP SYN-ACKs and breaking SSH access.

Resolution: Injected `install_routes = no` into `strongswan.conf` via Azure Run Command and executed `ip route flush table 220`, forcing XFRM to respect the native system routing table.

2. IKE_AUTH Payload Failure (Phase 1 MAC Mismatch)

Diagnosis: The Azure VNG rejected the initial handshake. Logs displayed `tried 1 shared key for '%any'... but MAC mismatched` due to a static ID string mismatch.

Resolution: Converted `/etc/ipsec.secrets` to utilize wildcard matching syntax (`: PSK "YourKey"`) to safely bypass the Azure endpoint's strict ID verification.

3. Traffic Selector Alignment (Phase 2 ESP Drops)

Diagnosis: Phase 1 (IKE) and Phase 2 (Child SA) reported ESTABLISHED, but end-to-end ping/SSH failed.

Resolution: Identified the Azure `rightsubnet` was configured as a spoke (`10.1.0.0/16`). Updated to match the actual Hub VNet (`10.0.0.0/16`). The strict kernel policy immediately began encrypting payloads successfully.

5. Final Validation Output

```
User@strongswan:~$ sudo ipsec statusall
```

```
Security Associations (1 up, 0 connecting):
```

```
azure-s2s[2]: ESTABLISHED 12 minutes ago, 192.168.254.4...<AZURE_VNG_PIP>
```

```
azure-s2s[2]: IKEv2 SPIs: fa6a3fc63d16a5c4_i* 4563459e49026a89_r
```

```
azure-s2s[2]: IKE proposal: AES_CBC_256/HMAC_SHA1_96/PRF_HMAC_SHA1/MODP_1024
```

```
azure-s2s{3}: INSTALLED, TUNNEL, reqid 1, ESP in UDP SPIs: c591dd4d_i  
451fbf2c_o
```

```
azure-s2s{3}: AES_CBC_256/HMAC_SHA1_96, 51974 bytes_i (514 pkts, 19s ago),  
50078 bytes_o
```

```
azure-s2s{3}: 192.168.0.0/16 === 10.0.0.0/16
```